

Segregation can occur anywhere that the flow of material is incorrect.

If a segregation stripe becomes apparent in the surface of the bituminous layer, follow the stripe to its source. Observe the action of the head of material at that point. Verify the material is flowing in a uniform manner at the correct speed and volume. Any interruption to the normal flow characteristics can cause larger aggregates to separate and form stripes or patches.

For example, in the above illustration the material is flowing forward around the tractor mainframe instead of flowing out to the end gate. As the larger aggregates flow down the face of the pile, they separate and form a stripe. In this case, there is a simple explanation. At the start of this paving pass, the paving width was 5 meters (16.5'). Gradually the paving width tapered to 3.8 meters (12.5'). As the screed extensions were moved in to create the specified paving width, material was trapped between the end gate and the end of the main screed. The material in front of the screed extension became slightly compacted and formed a barrier to the loose material coming out of the auger chamber. Instead of flowing out to the end gate and under the screed extension, the material is rolling forward and segregating. There is a simple solution to this problem. Remove the compacted material in front of the extension until the area is filled with fresh, loose material that moves out to the end gate.

[TROUBLESHOOTING GUIDE]

Use the following troubleshooting guide to help determine the cause(s) of common defects in the bituminous layer. There may be more than one factor contributing to a defect. When troubleshooting any problem, be methodical and change one condition at a time.

SURFACE TEXTURE

Problem	Remedy
1. Open texture behind extension	- Increase extension angle of attack - Increase heat; check screed heat system for malfunction - Increase depth to prevent dragging
2. Shiny texture behind extension	- Decrease extension angle of attack
3. Texture stripes, continuous	- Flatten trailing edge of screed Increase heat; check screed heat - Inspect screed for damage or wear
4. Texture Stripes, intermittent	- Check for high spots in grade; correct grade defects; clean up spills - Increase depth
5. Open texture full width	- Raise tow-point to correct line of pull - Increase angle of attack at take-off - Reduce head of material to prevent overreaction by grade control system - Increase plant output temperature - Activate vibratory system - Increase vibratory frequency - Reduce paving speed - Increase screed heat temperature - Increase layer thickness
6. Shiny texture full width	- Lower tow-point to correct line of pull - Decrease angle of attack at take-off - Decrease vibratory frequency
7. Open texture either side of center	- Increase auger height - Decrease lead crown
8. Open texture in center	- Increase lead crown - Check condition of reversing augers - Clean deflector plates

SEGREGATION

Problem	Remedy
1. Segregation one side of layer	- Look for segregation in storage silo - Check opening of clam gates in silo - Inspect loads in haul unit bodies - Load haul units from opposite direction
2. Stripe at longitudinal joint	- Reduce overlap onto cold side - Adjust end gate height - Adjust notch height if using notched wedge joint
3. Continuous stripes	- Adjust auger speed to 20-40 rpm - Check for high spots in the grade - Check for worn or damaged augers - Check for trapped material - Add auger / mainframe extensions
4. Intermittent stripes	- Check for erratic auger speed - Adjust auger speed to 20-40 rpm - Clean up spills in front of paver - Check for high spots in the grade
5. Stripe in the center	- Clean deflector plates - Check reversing augers / paddles - Add lead crown
6. Repetitive patches	- Check haul unit loading at plant - Inspect for segregation in truck bodies - Keep hopper at least half full - Stop folding hopper wings - Fold hopper wings over full conveyors - Increase windrow overlap
7. Random patches	- Inspect material coming out of silos - Check for on / off auger operation - Adjust auger speed to 20-40 rpm - Stop paving before emptying hopper Insert - Stop paving before emptying transfer vehicle

ROUGHNESS

Problem	Remedy
1. Ripples	- Adjust feeder controls to get consisten head of material - Reduce head of material - Check for excessive auger wear - Check for worn screed plates - Adjust paving speed to be consistent - Reduce haul unit brake pressure - Check for variable material temperature; correct at asphalt plant - Check for excessive play at screed pivot point - Check for excessive play at thickness screws - Increase tamper bar frequency or reduce paving speed to achieve correct tamper overlap - Adjust tow-point height to get parallel line of pull - Adjust angle of attack to get correct 3 mm (0.125") nose-up attitude
2. Wavy surface (short)	- Avoid over-correction with manual depth screws - Calibrate grade control system - Adjust feeder controls to deliver consistent head of material - Adjust auger speed to 20-40 rpm - Reduce head of material - Position grade sensor closer to tow-point
3. Wavy surface (long)	- Check grade reference for long depressions or high spots - Reduce time stopped waiting for trucks; reduce paving speed - Activate screed counterbalance system - Increase pressure in screed counterbalance system - Activate screed hold system, if equipped - Stop paving with conveyors full - Stop folding hopper wings if segregation is occurring
4. Intermittent roughness	- Calibrate grade control system - Inspect grade sensor(s) - Monitor averaging ski operation while paving super elevations - Clean spills in front of paver - Clean spills in front of grade sensor(s) - Correct high spots in grade prior to paving - Monitor feeder system operation - Discontinue automatic slope control; control slope manually - Install averaging ski in place of single grade sensor

BLEMISHES

Problem	Remedy
1. Drag marks	- Clean spills in front of paver - Correct high spots prior to paving - Increase layer thickness - Check slope of grade prior to using automatic slope control - Monitor head of material; do not run low - Increase screed heat
2. Oversized material	- Check gradation screens at plant - Check scalping screens at recycle stockpile - Clean truck bodies - Lower flow gates, if equipped
3. Screed marks	- Trucks stop short of push rollers - Reduce time stopped waiting for trucks; reduce paving speed - Activate screed counterbalance system - Increase pressure in screed counterbalance system - Activate screed hold system, if equipped
4. Rich spots / bleeding	- Correct moisture in mixture - Reduce asphalt cement content - Reduce vibratory frequency - Switch to static compaction - Plant maintenance to eliminate dust balls
5. Separation marks	- Adjust screed extension height - Adjust screed extension slope - Adjust slope stop to flatten screed extension

[SUMMARY]

New technologies have simplified the asphalt paving process and made it easier for crews to optimize paver performance, productivity and efficiency. Yet the human element remains as the single, most important factor in utilizing pavers to their maximum advantage.

Many worksite circumstances can arise that result in problems laying down a high-quality mat and/or meeting daily production goals. There is no substitute for crewmember experience in preventing problems, and when they do occur, quickly solving problems.

Cat Paving Operations Training is one option that pays immediate dividends and much more over time. This *Guide to Asphalt Paving* is another key tool that should be available for ready reference. Unit 8 is a great guide to diagnosing and solving problems. Many of the topics may sound familiar, but even the most experienced supervisors, paver and screed operators, and quality personnel are likely to learn something new and gain a deeper understanding of many common defects, some visible, some not.

The surface texture of the bituminous layer should be uniform, edge-to-edge, across the full width of the layer. When problems appear in the mat, crewmembers need to know the possible causes, determine what actually happened, and immediately take corrective action. The photos and descriptions in Unit 8 will help with determining the cause.

Segregation can be a particularly annoying problem. Knowing where it is happening and what to do to eliminate or minimize its impact on the bituminous layer is critical. Guidelines for root cause analysis will help determine if the problem is originating at the plant, in the haul units, in a transfer device, in the paver hopper or hopper insert, or in the paver's auger chamber.

The concise Troubleshooting Guide at the end of this unit is a great resource for spotting problems and applying remedies involving surface texture, segregation, roughness and blemishes. Find a place to store this guide on the paver or in a worksite vehicle. When it comes to understanding what is happening on the worksites, knowledge is power. This guide presents decades of knowledge packed into an easy-to-use, illustrated format.

GLOSSARY OF TERMS

- A -

Aggregate	Aggregate refers to the type or types of stones used in the manufacture of bituminous paving material.
Crushed Stone	The mechanical reduction of gravel and rock by crushing machines. Crushed stone is graded by running through vibrating screens.
Gravel	Large granular material produced by the natural erosion or disintegration of rock. Gravel usually has rounded surfaces and is larger than 6 mm (0.25").
Sand	Fine granular material resulting from the natural disintegration of rock or the crushing of rock. Often classed as smaller than 6 mm (0.25").
Moisture Content	Moisture content is an important factor regarding the suitability of an aggregate for use in bituminous material. Excessive moisture means increased drying time and less production. Excessive moisture in the mix creates instability.
Air Voids	Air voids are pockets of air trapped within an asphalt layer that has been laid down by a paving screed.
Ambient Conditions	Ambient conditions include air temperature, wind speed and direction, and cloud cover. Ambient conditions affect heat loss in the asphalt from the time of manufacturing at the plant until transfer into the paver.
Approach	An approach is the connection to an existing structure, such as an intersection or ramp, that is perpendicular to one side of the primary paving pass.
Asphalt	Asphalt is the generic term used to describe bituminous paving material, also called hot mix asphalt or sometimes warm mix asphalt. Asphalt can also be classified as cold mix.
Asphalt Cement	Asphalt cement is the oil used in the manufacture of bituminous paving material. Asphalt cement is also called bitumin.
Cutback	Cutbacks are asphalt cements mixed with a hydrocarbon solvent, typically a petroleum distillate, in order to lower the viscosity of the asphalt cement.
Doped	Asphalt cement with additives that improve adhesion.
Elastometric	Polymer-modified asphalt cement with higher elastic properties and less thermal susceptibility.
Fluxed	Asphalt cement that is softened by the addition of low-volatile fluxing oil.
Modified	Usually refers to polymer-modified asphalt cement with increased stiffness.
Pigmented	Asphalt cement that has had a coloring agent added.
Asphalt Concrete	Asphalt concrete is another phrase used to describe bituminous paving material.
Thin Lift	Asphalt wear course 30-50 mm (1.2-1.9") thick.
Very Thin Lift	Asphalt wear course less than 30 mm (1.2") thick.
Asphalt Plant, Batch	A batch plant produces a fixed quantity of mix in one discontinuous operation. That batch is transferred to a storage silo.
Asphalt Plant, Continuous	Also known as a drum mix or continuous plant, it is continuously fed aggregates, additives and asphalt cement. Finished material is transferred to one or more storage silos.
Auger	Continuous, rotating screws used to move asphalt across the left and right faces of the screed.
Height	The distance between the bottom of the auger segments and the surface of the bituminous layer. Typically, auger height is adjusted to 5 to 8 cm (2 to 3").

Speed	Auger speed is the rotation speed of the augers during the paving operation. Normally set at 20 to 40 revolutions per minute.
Chamber	The space between the rear of the tractor frame and the face of the screed.
Auger Shadow	Auger shadows are open-textured stripes that appear in the surface of the layer in line with the augers. The shadows are caused by the augers being set too low.
Averaging Ski - Mechanical	A mechanical averaging ski is towed by the paver and contacts the grade reference. May be installed outside or inside the paving width.
Averaging Ski - Electronic	An electronic averaging ski is attached to the paver and does not contact the grade reference. An electronic ski uses multiple sonic grade sensors and averages the signals. May be installed outside or inside the paving width.

- B -

Baghouse	Wet or dry dust collection systems that are part of the air-quality-control system at an asphalt plant. Baghouse dust is commonly recycled as part of the asphalt mixture.
Base Layer	The base layer is usually the first layer of bituminous material in a roadway structure. The base layer usually consists of large aggregates and is laid 75 mm (3") thick or thicker.
Binder	Organic material (bitumen, coal tar) with properties of cohesion and adhesion that enable it to bond aggregates.
Binder Layer	The binder layer is bituminous material laid on top of the base layer. The binder layer consists of medium size aggregates and is usually 50 -100 mm (2 -4") thick.
Binder Content	Expressed as a percentage, binder content is the ratio between the weight of the binder and the weight of dry aggregates in the mix. Or, it is the ratio between the weight of the binder and total weight of the mix.
Bitumen	Bitumen is a black, sticky mixture of hydrocarbons found naturally or obtained as a residue from petroleum distillation. Bitumen is also called asphalt cement.
Modified	Modified bitumens contain additives such as polymers that improve the bitumen's engineering properties.
Straight	Straight-run residue resulting from atmospheric distillation of petroleum crude oil.
Performace Grade	The performance grade system classifies bitumens based on engineering properties at various expected high and low pavement temperatures.
Bituminous Material	Bituminous material is the combination of aggregates, asphalt cement, and certain additives. Bituminous material is manufactured at asphalt plants.
Hot Mix	Bituminous material produced in an asphalt plant at temperatures between 150 – 180° C (300 – 350° F).
Warm Mix	Bituminous material produced in an asphalt plant at temperatures up to 40° C (100° F) lower than hot mix.
Cold Mix	Mixture of aggregates and binders that provides coating without heating and drying the aggregates. Bitumen emulsion is typically used as binder.
Bleeding	The undesirable effect of asphalt cement rising to the surface of a wear course.
Braking Force Coefficient	The coefficient of braking force is a measurement that characterizes skid resistance of the longitudinal roadway surface.
Bulk Density	Density of loose, or non-compacted bituminous material.

Butt Joint — A butt joint (also called a night joint) is a common term used to describe the transverse starting point for a paving pass that joins a previously laid pavement section with a new section.

- C -

Chamfer — A chamfer is the slope created by beveling the edge of the pavement, usually at a 45° angle. A chamfered edge resists deformation and the effects of weather.

Clam Gates — Clam-type discharge gates are designed to open simultaneously from the center and discharge a mass of hot bituminous material into a storage silo. Improper gate opening can lead to segregation in the storage silo.

Clean-out Area — An area specified for clean-out of haul units. May be on the project or at the asphalt plant.

Compaction — Compaction is the mechanical process of reducing air voids and developing load-bearing strength in a layer of bituminous material by moving aggregates into interlocking contact.

Compaction Rate — The difference in thickness between the layer passing under the screed (pre-compaction height) and the compacted thickness after the compaction process is complete.

Confined Edge — A confined edge is an edge of an asphalt layer that is confined by a previously laid layer. The intersection of the two layers is called a longitudinal joint.

Control House — The control station for the asphalt plant operator that contains all the controls for production, storage and discharge of bituminous material.

Conveyors — Located in the paver hopper, left and right slat conveyors move material from the hopper to the auger chamber in front of the screed.

Cooling Curve — A cooling curve is a graphical time chart of the heat loss in an asphalt layer based on layer thickness, type of material and ambient conditions.

Core Sample — A core sample is a small portion of the cooled, compacted asphalt layer that is removed by quality control personnel and taken to a laboratory for quality analysis.

Core Temperature — Core temperature refers to the temperature measured by a probe inserted into the center of an asphalt layer. Core temperature is always higher than surface temperature and is the true indicator of an asphalt layer's workability.

Corrugations — Corrugations, or washboarding, are a form of plastic deformation that results in ripples across the pavement surface. Usually caused by repeated vehicle braking at intersections.

Crack Seal — The use of an asphalt emulsion or similar product to fill cracks in the pavement and prevent moisture penetration. Depending on the type and age of the crack sealant, it may expand and cause bumps when covered with hot asphalt.

Cracking — Various types of defects, usually visible in the surface layer.

- **Fatigue** — Series of interconnected cracks caused by fatigue failure of the surface under repeated traffic loading.
- **Longitudinal** — A crack or discontinuity in a pavement that runs generally parallel to the pavement centerline. Longitudinal cracks may occur as a result of poorly constructed paving lane joints, thermal shrinkage, inadequate support, reflection from underlying layers, or as a precursor to fatigue cracking. Longitudinal cracking that occurs in the wheel path is generally indicative of alligator (fatigue) cracking.
- **Reflective** — Reflective cracks occur in overlays and reflect the crack pattern in the underlying layer.
- **Slippage** — Slippage cracks are crescent shaped cracks resulting from traffic-induced horizontal forces. Typically caused by a lack of bonding between the surface layer and the binder layer.

Transverse	Transverse cracks occur at approximately right angles to the centerline of the structure and often at regular intervals. The cause is movement due to temperature changes and hardening of the asphalt cement with aging.
Counterbalance System	The counterbalance system, when activated, puts a certain amount of upward hydraulic pressure in the screed lift cylinders. Screed counterbalance takes some weight off the screed and transfers that weight to the tractor. Used to minimize screed marks in the layer when the paver stops.
Cross Coupling	Cross coupling is a feature of some grade and slope control systems that ties together the movements of both tow-points to provide simultaneous grade and slope corrections.
Cross Section	A cross section is an engineering drawing that illustrates the transverse profile of a section of pavement at one location.
Crown - Structural	The crown of a structure is the intersection of two layers with opposing slopes.
Crown - Screed	The main screed plate may be set at the midpoint to create positive or negative opposing slopes.
Cutoff Shoe	A cutoff shoe is installed in the auger chamber to reduce the paving width to a measurement less than the width of the main screed.
Cycle Time	Cycle time is the time it takes for a haul unit to make a round trip between the asphalt plant and the project. Cycle time is the critical element in determining the number of haul units required to match plant production.

- D -

Delamination	The separation of two layers of bituminous material. Can be caused by insufficient tack between the layers or by the cold planing process.
Density	Density is the weight of a given volume of material, normally expressed as kilograms per cubic meter or pounds per cubic foot.
Density Testing Gauge	Density testing gauges are used to test density on the project while the bituminous material is still hot enough to make adjustments. The quality control technician calibrates and uses the gauge on the project.
Density - Theoretical Maximum	Theoretical maximum density is the weight of a given volume of bituminous material compacted in a repeatable, controlled manner in a laboratory environment.
Density Required	Required density is expressed as a percentage and represents the ratio between the weight of a sample of the compacted mix and the weight of a laboratory-compacted reference sample (Marshall, Duriez, for example).
Depth Crank	Depth cranks are located on each side of vibratory screeds. Layer thickness can be changed manually through the depth cranks.
Design Ratio	Design ratio is defined as the ratio between the thickness of asphalt layer and the largest aggregate in the layer. Most public works departments require at least 3:1 ratio of layer thickness to aggregate size. The design ratio has a direct effect on the compaction process. A higher ratio is better.
Drag Marks	Drag marks are open-textured stripes in the surface of the bituminous layer. They can be caused by a cold screed, a damaged screed, insufficient angle of attack, or layer not thick enough to float the screed.

- E -

Eccentric Weight	The eccentric weight is an off-center mass on a vibratory shaft mounted on the paving screed. Rapid rotation of the eccentric weight creates forces that cause the screed to vibrate and to transmit vibratory waves into the asphalt layer.

Echelon Paving — Echelon paving is the use of two or more pavers to lay down bituminous material while creating a hot longitudinal joint.

Edge Cutter — An edge cutter is an attachment, frequently installed on a compactor drum for trimming the unconfined edge of an asphalt layer to provide a vertical face and straight line for joint matching.

Effective Working Speed — The effective working speed is the actual working speed multiplied by the efficiency factor.

Efficiency Factor — An efficiency factor should be used when calculating the working speed of an paver to compensate for stop, waiting for trucks or for other reasons.

Elevation — In simple terms, elevation is the height of any pavement layer. Elevation is often expressed as the thickness of a given layer in relation to a surveyed grade.

End Gate — A paving screed has left and right end gates that confine the asphalt layer to the desired width. The bottom of an end gate has a metal strip called a ski. When the end gate is down, the ski floats on the grade and the edge of the asphalt layer is vertical for best joint matching.

Equilibrium — Equilibrium is the point at which all the factors affecting the screed are constant and the screed is floating in a relaxed condition at a constant level.

Extension Screed

- **Power** — An extender is the outer portion of the screed that can be moved in or out hydraulically to change the paving width. Screed extenders are in front of the main screed on some pavers and behind the main screed on others.
- **Bolt-on** — Bolt-on screed extensions are available in several different widths and are secured to the end of the power extenders for extra paving width.
- **Height** — The height of a screed extender is its vertical relationship to the main screed.
- **Slope** — A screed extender can have a positive or negative slope in relation to the main screed.

Extension

- **Auger** — Auger extensions are attached to the ends of the auger shafts to help move material out to end gates while maintaining an acceptable head of material.
- **Mainframe** — Mainframe extensions are bolted to the end of the tractor mainframe or may be hydraulically extendable. Mainframe extensions should always be added in front of auger extensions.

Extender Mark — An extender mark is a continuous, longitudinal line in the surface of the bituminous layer. Created by a height mismatch between the main screed and the extender.

- F -

Fat Spots — Dust clumps that appear in the bituminous layer. Their presence usually indicates that baghouse maintenance is needed.

Feed Bins

- **Cold** — Cold feed bins contain the aggregates used in a particular mix design. Each bin is loaded with one size aggregate. Conveyors move cold aggregates to the drum mixer or hot feed bins.
- **Hot** — Hot feed bins receive metered amounts of cold aggregates for heating prior to transfering them to the pugmill in a batch plant.

Feeder Sensors — Feeder sensors measure the level of the material at the end of the auger shafts or the level of material behind each conveyor in the auger chamber. There are sonic feeder sensors and mechanical feeder sensors.

Feeder System — The feeder system of a paver consists of left and right conveyors, left and right augers and feeder sensors.

Fines — The smallest particles in the mixture formula. Fines combine with asphalt cement to help create the binding effect between larger aggregate particles.

- **Aggregate Fines** — Aggregates that pass through the 4.75 mm (0.2") sieve opening.
- **Baghouse Fines** — Material added to the mix after dust collection in the baghouse.

Flow Gates — Flow gates are vertically adjustable strike-offs installed over the right and left conveyors at the back of the hopper. They provide one way to control the level of mix in the center of the auger chamber.

Foamed Asphalt — Mix that uses foamed (expanded) binder created by adding small amounts of water or steam to hot liquid asphalt.

Fore-'N-Aft Leveler — The Fore-'N-Aft Leveler is a mechanical averaging ski that is installed to extend over the screed and use a grade inside the paving width.

Frequency

- **Vibratory** — Vibratory frequency is the rotational speed, expressed as vibrations per minute, of the eccentric weight shaft installed on the paver screed.
- **Tamping** — Tamping frequency is the number of vertical strokes made by the tamper bar in one minute.

- G -

Gradation — Referring to bituminous material, gradation is a systematic progression in the size of the aggregates used in the mix design formula. A gradation curve indicates sizes and amounts of each aggregate to be used.

- **Continuous** — Term used to describe a gap-free gradation curve.
- **Dense-Graded** — Dense-graded mixes are produced with a variety of aggregate sizes, asphalt cement and fines. The larger aggregates are surrounded by a mastic of fines and asphalt cement.
- **Gap-Graded** — Gap-graded mixes use an aggregate gradation with particles ranging from large to fine with some intermediate sizes missing. Gap-graded mixes are permeable with lots of stone-on-stone contact.
- **Double Gap-Graded** — Bituminous mix with a double gap in its gradation curve.

Grade — Often refers to the base over which the bituminous material is being laid.

Grade Control — Maintaining the correct elevation of the bituminous material's surface. Also refers to controlling the thickness of the bituminous layer.

Grade Control Automatic — Using an on-board control system with sensors and computers to regulate the elevation of the bituminous layer's surface.

Grade Sensor — A grade sensor, either sonic or mechanical, is a measuring device that sends data to the system controller regarding the elevation of the grade reference.

Grade Stake — A grade stake is a construction marker erected adjacent to the pavement for use as an elevation and/or slope indicator.

Guidance Value — A reference number selected by a screed operator and shown on the display of the grade and slope control system.

- H -

Haul Units — The trucks utilized to transport the asphalt from the production facility to the worksite.

- **Bottom Dumps** — Dump from the bottom of a trailer in a windrow requiring a transfer device to feed the paver hopper.
- **End Dumps** — Dump from the rear of the truck or a trailer into a material transfer device or directly into the paver hopper.

Head of Material — The level of material in the auger chamber in front of the screed. A major influence on how the screed floats.

High Modulus Asphalt — Stiff bituminous mix that resists deformation under load. Typically specified for roadways carrying high traffic volume.

Hopper — The hopper is a compartment at the front of the paver tractor. It provides limited storage capacity while the paver is receiving material from the haul unit. The sides of the hopper can be folded to move material from the sides of the hopper onto the conveyors.

- **Flashing** — Hopper flashing helps contain material at the front of the hopper.
- **Folding Apron** — A folding apron is an option that helps move material from the front of the hopper back to the conveyors.
- **Insert** — A hopper insert increases the storage capacity of the hopper so the paver can operate for longer intervals between trucks.

- I -

Impact — Impact is a dynamic compaction force. Impact occurs when the tamper bar in front of the screed moves into the asphalt layer.

Impact Marks — Impact marks are visible lines in the surface of the asphalt layer. Impact marks are caused by applying too much force to the asphalt layer or incorrect impact spacing.

Impact Spacing — Impact spacing is the relationship between the frequency of a tamper bar and the working speed of an paver. Impact spacing should be such that there is at least 6 mm (0.25") overlap between tamper bar impacts.

Infrared Camera — An infrared camera produces an image of the asphalt layer's surface temperature. The image usually encompasses the width of the mat for a distance of no more than 9 meters (30').

Infrared Scanner — An infrared scanner is a hand-held device that measures and displays the temperature of a point at the surface of an asphalt layer.

Initial Phase Compaction — The initial phase of compaction occurs immediately behind the paver where the asphalt layer is the hottest. The initial phase should achieve the majority of the target final density.

Intermediate Phase Compaction — The intermediate phase of compaction occurs immediately after the initial phase in a temperature zone where the asphalt layer is still hot enough to allow for gains in density. The intermediate phase should achieve the final density target.

Internal Friction — Internal friction is the resistance to movement by the aggregate in an asphalt layer. The shape of the aggregate determines the amount of internal friction.

International Roughness Index — The International Roughness Index is a measure of ride quality. A certified test vehicle drives over a paved surface at normal highway speed and records chassis movement that is induced by roughness.

- L -

Lead Crown — Lead crown exists when the front of the main screed plate has a slight crown, 3 mm (0.125") for example, while the trailing edge of the screed plate is flat.

Level Section — A level section is a portion of a pavement structure with 0 percent slope.

Leveling Layer — A leveling layer is a thin layer of bituminous material laid on a milled surface for the dual purposes of restoring profile and improving smoothness.

Lifting Force — Lifting force is the hydrodynamic force created by the material passing under and supporting the floating screed. Lifting force is affected by type of mix, layer thickness and material temperature.

Line of Pull — Also called the traction angle, line of pull is the relationship between the height of the tow-point and the screed pivot point.

Load Cell	Scales on either side of storage silos that measure the empty and loaded weight of haul units.
Longitudinal Joint	A longitudinal joint is the intersection of two asphalt layers along the edges that are parallel to the direction of paving.
Hot/Cold Joint	A hot/cold longitudinal joint is created by laying down a hot asphalt layer adjacent to a cold, previously compacted asphalt layer.
Hot/Warm Joint	A hot/warm longitudinal joint is created by laying down a hot asphalt layer adjacent to a layer laid down a short time before.
Hot/Hot Joint	A hot/hot longitudinal joint is created by two pavers laying down adjacent mats in echelon.
Loose Thickness	The thickness of the bituminous layer laid by the screed prior to compaction.
Loose Weight	The weight of the bituminous material prior to compaction expressed in kilograms per cubic meter (pounds per cubic foot). Useful when calculating paving speeds.
Lower Lock - Screed	On pavers with tamping screeds, the screed lower lock engages when the paver is in the Standby mode to prevent screed marks while the paver is stopped on uncompacted bituminous material.

- M -

Mastic Asphalt	Mixture of bitumen, aggregate and fillers that is fluid when hot. Mixed in a stationary or mobile plant, it is transported in a mixing truck. Used on sidewalks or wear courses or bridge decks as a waterproofing agent, mastic asphalt is spread by hand.
Material Transfer Vehicle	Self-propelled machine that receives material from the haul units and transfers the material to the paver hopper or hopper insert. Makes continuous paving easier and may provide remixing to reduce risk of segmentation.
Mechanical Sensor	A mechanical sensor, either grade-type or feeder-type, contacts the reference target.
Micro-milling	Removal of a specified amount of pavement material by a cold planer equipped with a high-density cutting drum.
Milled Surface	The grade left after a specified amount of pavement material has been removed by a cold planer.
Mineral Filler	Also referred to as fines, mineral fillers can be crusher dust, dry Portland cement, fly ash or lime. Fillers help fill in the spaces between larger aggregates.
Mix Formula	The formula or mix design approved for the production of bituminous material on a specific project. It describes aggregate gradation, asphalt cement type and asphalt cement content.
Mix Height Dial	Potentiometers located at each screed control box used to adjust the head of material at the outboard end of the augers. Pavers equipped with four feeder sensors also have mix height dials at the operator's station to control the head of material at the end of the conveyors.
Mix Sample	A small portion of bituminous material that will be tested for conformance to design parameters. Samples are taken from various locations such as the plant discharge, the paver hopper, the paver auger chamber and the layer itself.
Mobilization	Mobilization is activity associated with transporting equipment and supplies to a project location. Mobilization may be a separate item on a bid request.
Mounting Hardware	Mounting hardware consists of the arms that support grade sensors and averaging skis. Mounting hardware is attached to the paver tow arms or the end gates.

- N -

Nose Bar	The nose bar is a rounded wear strip installed in front of the screed plate.
Notched Wedge Joint	A notched wedge joint is created by installing a joint shaper at the end of the paving screed. Notched wedge joints are often specified to avoid creating a large vertical, unconfined edge that may be open to traffic.
Null, Screed	Null the screed by turning depth cranks to remove any twist in the screed while the screed is resting on the starting point.
Null, Sensor	Nulling, also called benchmarking, a grade sensor or slope sensor creates a reference distance for a grade sensor or a reference slope for a slope sensor.

- O -

Offset Joint	An offset joint is created by varying the width of layers that form a longitudinal joint as the pavement structure is built. Offsetting or staggering joints avoids creating a series of joints in one location when placing multiple layers.
Open-Graded Mix	Open-graded bituminous material is made up of only a few aggregate sizes with intermediate sizes absent. Modified asphalt cement is usually part of the design. There is a lot of stone-on-stone contact.
Outboard Leveler	An outboard leveler is a mechanical averaging ski that uses a grade reference outside the paving width.
Overlap	Overlap refers to the distance the end gate ski extends onto the adjacent asphalt layer
Overlay	Placement of bituminous material on existing pavement.
Oversize	Oversize is any material discharged into the paver hopper that is larger than the maximum aggregate size specified by the mixture formula.
Oxidation	Hardening of asphalt cement after exposure to oxygen over a long period of time.

- P -

Pass	A pass is one movement of the paver in one direction from the starting point to the point where the screed is picked up and paving stops.
Paving Train	The paving and compaction equipment assigned to a worksite from asphalt delivery through the compacted, finished surface.
Penetration Test	Test used to measure the hardness of bitumens. Based on hardness, bitumens are classed into various grades.
Permeable Asphalt	Also referred to as "draining asphalt," these mixes have a high void content enabling water to flow through in the connecting voids.
Perpetual Pavement	Perpetual pavement is a term used to describe a full-depth asphalt structure designed to withstand an almost infinite number of axle loads without structural deterioration.
Pivot Point	The screed pivot point is the point on the screed from which the angle of attack is generated.
Planing Angle	The attitude at which the screed floats on the bituminous layer. Also referred to as the angle of attack.
Polymer Modified Asphalt	A polymer is a synthetic compound in the form of a string of similar-linked molecules. Polymers are added to asphalt cement to enhance layer strength at high temperatures and layer elasticity at low temperatures. Polymer-modified asphalt cement has high viscosity.
Pre-Compaction	Pre-compaction is the amount of density imparted to the asphalt layer by the screed. Pre-compaction rates are affected by screed weight, paving speed, and the energy delivered by a vibrating and/or tamping screed.

Profile — Profile is the cross section of a roadway structure, most often referring to the slope of the surface of the roadway for drainage.

- **Transverse** — Another way to refer to the slope of the structure.
- **Longitudinal** — Longitudinal section of a roadway that shows changes in elevation over a given distance.

Profile Index — Profile Index is a trace of the surface of a pavement structure. It is generated by pushing a profilograph over the surface to be measured or by driving an inertial profiler over the surface.

Proof Rolling — Confirmation of the stability of an aggregate or reclaimed base prior to paving. Commonly done with heavy pneumatic rollers.

Pugmill — A pugmill consists of a rotating shaft, or a series of shafts, with paddles used to blend aggregates and asphalt cement in a batch plant.

Push Roller — Push rollers are located at the front of the tractor. They contact a haul unit's tires when the paver is pushing the haul unit that is dumping its load into the hopper.

- Q -

Quality Anaysis — Quality analysis is testing, measuring and analyzing bituminous material and other specified aspects of a project in a laboratory or other controlled environment.

Quality Control — Quality control is testing, measuring and analyzing bituminous material and other specified aspects of a project at the worksite while work is in progress.

- R -

Ratio Control Dials — Potentiometers located at the operator's station are used to adjust the speed of the conveyors. Potentiometers provide one way to control the level of material in the center of the auger chamber.

Reaction, Screed — Screed reaction is the time it takes the screed to change elevation in response to a manual depth correction or to a correction sent by the automatic grade control system.

Reclaimed Asphalt Pavement — Reclaimed asphalt pavement, also called milled material, is the material removed from an existing asphalt layer. Reclaimed asphalt is often specified as part of a bituminous material formula.

Release Agent — A release agent is some type of liquid that helps prevent bituminous material from sticking to steel or rubber surfaces. Petroleum distillates, like diesel fuel, are prohibited in most locations because of their harmful effects on bituminous material. There are a variety of biodegradable release agents.

Reversing Augers — Reversing auger segments, or kicker paddles, are installed on the inner ends of the auger shafts. They move material under the center chain case to eliminate or minimize segregation in the center of the bituminous layer.

Rich Spot — A rich spot in the bituminous layer is an accumulation of asphalt cement and fines or dust.

Ripples — Ripples are closely spaced deviations or bumps in the surface of the bituminous layer. Also called corrugations or washboarding.

Rock — Naturally occurring minerals.

- **Igneous** — The result of cooling and hardening of molten materials beneath the surface of the earth. Igneous rock is a durable aggregate, but can cause accelerated wear on paver parts.
- **Metamorphic** — Formed by exposing sedimentary or igneous rocks to prolonged heat and pressure. Metamorphic rocks are very hard, but are brittle and can be damaged by heavy impact forces.
- **Sedimentary** — Formed from sediment collecting in water or wind-blown deposits such as limestone and sandstone. Can be fragile and easily damaged during paving and compaction.

Roller-Compacted Concrete Roller-compacted concrete is a hydraulically bound cement material used for pavement layers and as base support layers. It gains strength through compaction and chemical curing.

Rolling Resistance Rolling resistance is the force that works against the motion of an object on a surface. For example, a smooth pavement surface offers less rolling resistance than a rough pavement surface.

- S -

Screed That portion of the paver that is towed by the tractor and establishes the layer width, layer depth and pre-compaction density.

- **Extendable** Screed that includes power extendable extensions used to vary the paving width.
- **Fixed Width** Screed that consists of a main screed of a given width. Hard extensions are bolted on to vary the paving width.

Screed Imprint A screed imprint, also called screed settlement mark, is the indentation in the hot bituminous layer left by the screed during a paver stop.

Segregation Segregation refers to the tendency of larger aggregates in a bituminous mixture to separate from the smaller particles and form pockets or stripes of large aggregates within the layer.

- **Stripes** Stripe segregation is a continuous line of large aggregates that appear in the surface of the bituminous layer.
- **Patch, Pattern** Pattern patches of segregated aggregates appear in a regular manner and are commonly associated with end-of-load segregation.
- **Patch, Random** Random patch segregation is the appearance of pockets of large aggregates at irregular intervals.

Self Leveling The action of a floating screed that reduces high spots and fills in low spots while paving.

Separation Marks Separation marks are continuous lines in the surface caused by height mismatch between the main screed and a screed extender.

Set-Back A set-back is movement of the paver back to the start of the pass that was just completed in preparation for paving a matching pass.

Shear Force The shear force is the screed's ability to pass a certain amount of material under the front of the screed while pushing the remainder of the material in the auger chamber.

Sizing Screen Part of the aggregate feeder system at the plant, sizing screens help keep oversize material out of the dryer / mixer.

Slack As related to an paver, slack refers to lack of tension in the tow arm connections and screed connections. Remove the slack by pulling the paver forward a few centimeters while the screed is resting on the starting point.

Slag Slag is a nonmetallic byproduct of a blast furnace or a wet bottom boiler. Used as aggregate after testing and proper sizing. Can be porous and requires additional bitumen in the mix formula.

Slope The transverse inclination of the surface of the bituminous layer. Usually expressed as a percentage, slope is the change in elevation (rise) over a given distance (run).

Slope Board A slope board is a bubble-type level, usually 2.5 meters or 3 meters (8' or 10') long, used to measure the rise or fall (slope) of a pavement surface.

Slope Control Maintaining the lateral inclination of the paving screed to create the desired surface slope of the bituminous material.

Slope Control - Automatic Using a sensor and controller to maintain the lateral inclination of the paving screed.

Slope Meter A slope meter is an electronic level, usually 1 to 2 meters (3' to 6') long, used to measure the slope of a pavement surface.

Slope Sensor Installed on a transverse beam over the screed, the slope sensor measures the inclination of the screed and sends data to the system controller.

Slope Stop The slope stop is the limiter for the slope of a screed extender.

Specification - End Result An end result specification is a written quality control / quality analysis series of measurement targets for items such as ride quality, density and asphalt mix conformance.

Specification - Method A method specification describes either the type of equipment and/or the technique(s) that must be used on a project.

Starter Boards Starter boards are wood or metal strips placed under the paver screed before the screed is lowered to the starting point. Starter boards should be the same thickness as the compaction rate of the asphalt layer being laid down.

Stone Matrix Asphalt Stone matrix asphalt is composed almost entirely of large aggregates, fines and modified asphalt cement. There is stone-on-stone contact, but the stones are coated with a thick mastic of fines and viscous asphalt cement.

Stockpile Supply of one aggregate size at the asphalt manufacturing facility.

Strike-off Screed Some screeds have a vertically adjustable strike-off located on the face of the screed. Strike-off position affects how material flows under the screed nose bar.

Strike-off Extension Some screeds have a strike-off extension that is an integral part of the screed or one that is bolted onto the screed. Strike-offs are a leveling device and do not have a screed plate.

Stringline A stringline is a mechanical grade reference based on survey data and erected on one side of the paver for precise elevation control.

Super-elevation A super-elevation is a segment of the pavement with increased transverse profile (slope), usually designed to overcome the effect of centrifugal force on vehicles traveling through curved sections.

- **Runoff** A super-elevation runoff is the distance between a level section of pavement and a super-elevated section of pavement.

Surge Bin A storage unit, or silo, used to store bituminous material prior to discharge into haul units at the asphalt manufacturing facility.

Surge Capacity Surge capacity is the storage volume of a material transfer device or the paver hopper or hopper insert.

- T -

Tack Tack is an emulsion consisting of paving grade oil, water and an emulsifying agent. Tack is applied to surfaces prior to paving to help improve the bond between layers.

Tamping Screed A tamping screed uses one or more tamper bars to deliver extra compaction energy to the asphalt layer, thereby producing higher density in the asphalt layer prior to the compaction process. Tamping screeds typically use vibration to help tighten surface texture.

- **Amplitude** The distance that the tamper bar moves into the asphalt layer.
- **Dosing Angle** The angle on the face of the tamper bar.
- **Dosing Area** The length of the angled portion of the tamper bar.
- **Dosing Height** The distance from the top of the angled face to the tamper foot.

Tamping Foot The bottom of the tamper bar that contacts the surface of the bituminous layer.

Tangent Runoff Tangent runoff is the distance between a crown section of pavement and a level section of pavement. Commonly built at either end of a super-elevation.

Temperature Compensation Sonic grade sensors have some sort of temperature compensation to maintain accuracy when operating in conditions where temperatures may change rapidly.

Test Strip — A test strip is used to confirm that the proposed rolling pattern and equipment are adequate to achieve target density and match the estimated paving production. The test strip may be part of the project or may be a separate element.

Texture - Surface — Surface texture is the appearance of the surface of the bituminous layer. Texture is affected primarily by the screed angle of attack.

Thickness Screws — Thickness screws on each side of tamping screeds are adjusted to set the correct angle of attack.

Three Dimensional Paving — A screed guidance system that receives input from sources that are independent of the paver.

Tie-in — A tie-in is the connection to an existing structure, like a bridge deck that is perpendicular to the direction of paving.

Tow Arms — The tow arms on an paver are the connection between the screed and the tractor, which provides the power to tow the screed.

Tow-Points — The tow-points are the location where the tow arms connect to the tractor.

- **Cylinder** — The tow-point cylinder raises and lowers the tow-point connection to control tow-point height.
- **Height** — The tow-point height affects the line of pull and the screed angle of attack, or planing angle.

Transducer — A transducer is that part of a sonic sensor that emits sound pulses and uses the returning echoes to measure the distance between the transducer and the reference target.

Transition — Transition is the change in transverse profile between crown sections, level sections and super-elevated sections.

Transverse Joint — A transverse joint, also called a butt joint, is the perpendicular intersection of two layers of asphalt. Often a transverse joint is the starting point for resumption of paving from a cold, compacted layer.

Transverse Profile — Transverse profile is the slope of a pavement section.

Traffic Control — Traffic control is the activity associated with changing the flow of traffic to accommodate all construction activity, usually within specified times and according to local safety regulations.

Truck Exchange — A truck exchange is the procedure to replace an empty haul unit with a loaded haul unit in front of the paver or material transfer device.

Truck Guide — The truck guide is the person who is responsible for helping direct the flow of haul units in front of the paver and for directing the discharge of material from the haul units into the paver hopper or a material transfer vehicle.

Truck Hitch — The truck hitch is an option on xspavers. Truck hitch arms extend ahead of the push rollers to contact the haul unit's wheels and provide a solid link between the paver and the haul unit.

- U -

Unconfined Edge — An unconfined edge is the edge of an asphalt layer that is open and not bound by an adjacent layer, curb or gutter.

Undercarriage — That portion of the propel system that provides traction between the paver and the surface of the grade.

- **Rubber Belt** — Continuous reinforced rubber belt, either smooth or cleated. Preferred when traction and flotation are critical. Also provides high travel speed and tight turning radius.
- **Rubber Tire** — Two pneumatic tires, sometimes with front wheel assist, propel the paver. Preferred for its smooth ride and high travel speed.
- **Steel Track** — Continuous track-type undercarriage. High traction, but low travel speed.

Universal Total Station — A Universal Total Station is part of the guidance system used in three-dimension paving applications. Each station sends input to the on-board control system based on a computer-generated model.

- V -

Vibratory Screed — A vibratory screed delivers vibratory force to the asphalt layer as the layer passes under the screed. Screed vibration increases the density of the asphalt layer by a small amount and also helps tighten the surface texture.

Viscosity — Viscosity refers to the flow rate of a liquid at given a temperature. The viscosity of asphalt cement used in bituminous material is affected by temperature and by the additives mixed with the asphalt cement. For paving purposes, the viscosity of the asphalt cement affects the way the screed floats and how screed adjustments are made.

- W -

Waviness — Waviness is characterized by widely spaced undulations in the surface of the bituminous layer.

Wear Layer — The wear layer, often called the friction or surface layer, is the final layer in the roadway structure. Usually the thinnest layer, it is designed to be the stiffest layer.

Wedge Joint — A wedge joint is a tapered edge that eliminates a vertical face on the unconfined edge that may be open to traffic. A wedge joint is formed by installing a shaper at the end of the paver screed.

Weight Ticket — Printed document given to the truck driver at the asphalt plant. Contains load number, time, gross weight, net weight and cumulative daily project totals.

Windrow Elevator — A windrow elevator is a transfer device that is attached to the front of a paver. An elevating slat conveyor picks up material that is left in a windrow by the bottom-dump haul units and transfers the material to the paver hopper.

Windrow Elevator Head — A windrow elevator pickup mechanism with elevator used to load bituminous material deposited by bottom dumps in a linear column at grade into a material transfer vehicle.

Windrow Paving — Windrow paving is the process where the bituminous material is laid in a windrow by the haul unit on the base to be paved and then transferred to the paver hopper by a windrow elevator.

- Y -

Yield — Yield refers to the linear distance that a given amount of bituminous material will pave at a specified depth and width.

GLOSSARY


CAT


AP555F


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